

Class Identification: Tetris in Java Swing

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Class List:

UI Classes

Class 1: MainMenuPage

- Description: Displays the main menu to the user, with the different modes that are available including: Classic Mode, Score Attack Mode, Line Clear Challenge, VS AI Mode, Custom Mode, and High Score.

Class 2: ClassicModePage

- Description: Display a normal game of Tetris that implements the different mechanics.

Class 3: ClassicModeResultsPage

- Description: Displays the results after the user gets a game over for the classic mode, includes an option to save their score.

Class 4: ScoreAttackPage

- Description: Display the Score Attack gamemode.

Class 5: ScoreAttackPreviewPage

- Description: A page before the actual gamemode asking the user their time limit (3, 5, or 10 minutes)

Class 6: ScoreAttackResultsPage

- Description: After the user gets a game over or the time limit is reached, displays the user their score and saves their score into a file (uses the score saving class).

Class 7: LineClearingChallengePage

- Description: Displays the Line Clearing Challenge gamemode.

Class 8: LineClearingChallengePreviewPage

- Description: Displays the settings for the Line Clearing Challenge gamemode before the game begins.

Class 9: LineClearingChallengeResultsPage

- Description: Displays the results after the user finishes the Line Clearing Challenge.

Class 10: VersusAIPage

- Description: Displays the VS AI gamemode.

Class 11: VersusAIPreviewPage

- Description: Displays before the VS AI gamemode where the user can change the AI difficulty and chooses between Marathon Mode or Score Attack Mode.

Class 12: VersusAIResultsPage

- Description: Displays the results after the AI game which tells if the user wins or loses to the AI, includes an option to save their score.

Class 13: CustomModePage

- Description: Displays the Custom gamemode.

Class 14: CustomGamePreviewPage

- Description: Displays the different customization options available to the user.

Class 15: CustomGameResultsPage

- Description: Shows up after the user gets a game over and displays the score.

Class 16: HighScoresPage

- Description: Shows the user high scores in different game modes and allows users to import their high score files.

Elements/Mechanics Classes

Class 17: Startup (Entry Point)

- Description: Entry point of the program and starts at the main menu.

Class 18: Gameboard

- Description: Outlines the grid of the game, which can detect collisions with other blocks and whether the game is over.

Class 19: GameEngine

- Description: Handles timer (for other gamemodes), moves pieces with gravity, piece rotation, and hold mechanic.

Class 20: Tetriminoes

- Description: Stores all the different tetris pieces that are used in the program.

Class 21: 7Bag

- Description: An algorithm that randomizes all the Tetris pieces.

Class 22: Controls

- Description: Stores the controls for the program, this program will use keyboard controls.

Class 23: ScoreCalculator

- Description: A system that outlines the different ways to increase your score within a game.

Class 24: ScoreSaving

- Description: A system that will save your score into a file so you can import the file into the high score page to view.

Class 25: AI

- Description: Builds the AI that is used for the AI VS gamemode.

Class Relationships:

See google sheet for details

Key:

Row <relation to> column

Aggreg: aggregation (uses/depends)

Comp: compositions (has-a, not shared)

Inh: inheritance (is-a)

Class Diagram Framework

UI Classes

MainMenuPage
+ MainMenuPage:void (Constructor) + MainMenuGUI:void (Inner Class) + MainMenuEvents:void (Function within inner class)
Displays the main menu to the user, with the different modes that are available including: Classic Mode, Score Attack Mode, Line Clear Challenge, VS AI Mode, Custom Mode, and High Score.

ClassicModePage
+ ClassicModePage:void (Constructor) + ClassicModeGUI:void (Inner Class) + ClassicModeEvents:void (Function within inner class)
Display a normal game of Tetris that implements the different mechanics.

ClassicModeResultsPage
+ ClassicModeResultsPage:void (Constructor) + ClassicModeResultsGUI:void (Inner Class) + ClassicModeResultsEvents:void (Function within inner class)
Displays the results after the user gets a game over for the classic mode, including an option to save their score.

ScoreAttackPage
+ ScoreAttackPage:void (Constructor) + ScoreAttackGUI:void (Inner Class) + ScoreAttackEvents:void (Function within inner class)
Display the Score Attack gamemode.

ScoreAttackPreviewPage
+ ScoreAttackPreviewPage:void (Constructor) + ScoreAttackPreviewGUI:void (Inner Class) + ScoreAttackPreviewEvents:void (Function within inner class)
A page before the actual gamemode asking the user their time limit (3, 5, or 10 minutes)

ScoreAttackResultsPage
+ ScoreAttackResultsPage:void (Constructor) + ScoreAttackResultsGUI:void (Inner Class) + ScoreAttackResultsEvents:void (Function within inner class)
After the user gets a game over or the time limit is reached, displays the user their score and saves their score into a file (uses the score saving class).

VersusAIPage
+ VersusAIPage:void (Constructor) + VersusAIGUI:void (Inner Class) + VersusAIEvents:void (Function within inner class)
Displays the VS AI gamemode.

VersusAIPreviewPage
+ VersusAIPreviewPage:void (Constructor) + VersusAIPreviewGUI:void (Inner Class) + VersusAIPreviewEvents:void (Function within inner class)
Displays before the VS AI gamemode where the user can change the AI difficulty and chooses between Marathon Mode or Score Attack Mode.

VersusAIResultsPage
+ VersusAIResultsPage:void (Constructor) + VersusAIResultsGUI:void (Inner Class) + VersusAIResultsEvents:void (Function within inner class)
Displays the results after the AI game which tells if the user wins or loses to the AI, includes an option to save their score.

CustomModePage
+ CustomModePage:void (Constructor) + CustomModeGUI:void (Inner Class) + CustomModePageEvents:void (Function within inner class)
Displays the Custom gamemode.

CustomGamePreviewPage
+ CustomModePreviewPage:void (Constructor) + CustomModePreviewGUI:void (Inner Class) + CustomModePreviewEvents:void (Function within inner class)
Displays the different customization options available to the user.

CustomGameResultsPage
+ CustomModeResultsPage:void (Constructor) + CustomModeResultsGUI:void (Inner Class) + CustomModeResultsEvents:void (Function within inner class)
Shows up after the user gets a game over and displays the score.

LineClearingChallengePage
+ LineClearingChallengePage:void (Constructor) + LineClearingChallengeGUI:void (Inner Class) + LineClearingChallengeEvents:void (Function within inner class)
Displays the Line Clearing Challenge gamemode.

LineClearingChallengePreviewPage
+ LineClearingChallengePreviewPage:void (Constructor) + LineClearingChallengePreviewGUI:void (Inner Class) + LineClearingChallengePreviewEvents:void (Function within inner class)
Displays the settings for the Line Clearing Challenge gamemode before the game begins.

LineClearingChallengeResultsPage
+ LineClearingChallengeResultsPage:void (Constructor) + LineClearingChallengeResultsGUI:void (Inner Class) + LineClearingChallengeResultsEvents:void (Function within inner class)
Displays the results after the user finishes the Line Clearing Challenge.

HighScoresPage
+ HighScoresPage:void (Constructor) + HighScoresGUI:void (Inner Class) + HighScoresEvents:void (Function within inner class)
Shows the user high scores in different game modes and allows users to import their high score files.

Elements/Mechanics Classes

Startup
+ main:void
Entry point of the program and starts at the main menu.

Gameboard
Outlines the grid of the game, which can detect collisions with other blocks and whether the game is over.

GameEngine
Handles timer (for other gamemodes), moves pieces with gravity, piece rotation, and hold mechanic.

Tetriminoes
Stores all the different tetris pieces that are used in the program.

7 Bag
An algorithm that randomizes all the Tetris pieces.

Controls
Stores the controls for the program, this program will use keyboard controls.

ScoreCalculator
A system that outlines the different ways to increase your score within a game.

ScoreSaving
A system that will save your score into a file so you can import the file into the high score page to view.

AI
Builds the AI that is used for the AI Versus gamemode.